Exercises

(1) A bookseller had 800 copies of a novel. He sold 360 copies for Rs. 200 more than their cost. If he gained 20% on the whole by selling the remaining copies for Rs. 2800, find the cost price of a copy.

[*Ans* : Rs. 2.75]

(2) X sells his goods 15% cheaper than Y and 15% dearer than Z. A person purchases goods worth Rs. 1700 from X. If the person purchases half the goods from Y and half from Z does he stand to gain or lose, and by how much?

[*Ans* : No gain or loss]

(3) A shopkeeper sells an article so as to make 25% on the cost price. If he had sold it for 50 p. more, he would have made 30% profit. What is the selling price?

[*Ans* : Rs.10] [N.D.A.]

(4) A man bought a horse and a cart. If he sold the horse at 10% loss and the cart at 20% gain he would not lose anything. But if he sold the horse at 5% loss and the cart at 5% gain, he would lose Rs. 10 in the bargain. What did he pay for each?

[*Ans* : Horse : Rs.400; Cart : Rs. 200] [I.T.I.]

(5) In selling each of the two articles at Rs. 99, the dealer gains 10% in one, while he loses 10% in the other. Find his percentage gain or loss in the whole transaction.

[*Ans* : 1% loss]

(6) If the selling price of 5 articles is the same as the cost price of 4 articles of the same kind, find the percentage gain or loss.

[*Ans* : 20% loss]

(7) A person bought two articles, each at the same price. One of them he sold at a profit of 15% and the other at a loss of 10%. If the difference between the two selling prices is Rs. 6.25, find the cost price of each article.

[*Ans* : Rs. 25]

(8) A person sold 30 articles at a gain of 10% and 45 articles at a gain of 20%. Had he sold them all at a uniform gain of 20%, he would have received Rs. 3 more. Find the cost of each article.

[*Ans* : Re. 1]

(9) If a bookseller sells a book at the published price, he makes 40% profit on its cost. What percentage of profit will he make if he reduces the selling price of the book by 25%?

[*Ans* : 5%]

(10) The profit obtained by running a bus is proportional to the distance as well as the number of passengers exceeding a fixed number of them. When it travels a distance of 20 km. with 30 passengers, it fetches a profit of Rs. 30; while if it runs a distance of 35 km.

with 23 passengers, the profit is Rs. 28. Find the minimum number of passengers that should be taken in a bus so that running the bus will not result into a loss.

[*Ans* : 15]

(11) A dealer bought a motorcycle and a moped together for Rs. 7,200. He sold the motorcycle at 10% profit and the moped at 8% loss. If on the whole he earned 4% profit, find the cost price of each.

[*Ans* : Motorcycle : Rs. 4,800] [Moped : Rs. 2,400]

(12) 'A' sells a radio set to 'B' gaining 20%, 'B' sells it to 'C' gaining 10%. While 'C' sells it to 'D' for Rs. 880, thereby gaining 10% of the selling price. What was its cost to 'A'?

[*Ans* : 600]

(13) A dealer bought 30 tables at Rs. 50 each and sold them so as to make a profit which is 2/5th of the total cost. Find the percentage of profit and the selling price of each table.

[*Ans* : Profit 40% and S.P. Rs. 70]

(14) A radio set was being sold at a profit of 25%. The cost price as well as the selling price was increased by Rs. 100 each. If the profit in the new transaction was 20%, find the original cost price of the radio.

[*Ans* : Rs.400]

□□